Stat380 Final Report Group 3

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1. **Introduction**

   Our project focuses on the United States' market. We know that financial markets are sensitive to not only the contemporary economy state, but also how new information compares to what investors expected. The macroeconomic announcements like the Consumer Price Index (CPI), the unemployment rate, and Federal Reserve interest rate decisions are among the most closely monitored. These events are important because they can change expectations about inflation, economic growth, and future interest rates. When these pivotal changes are apparent, investors may adjust their portfolios, which can cause the stock market volatility to go either up or down.

   In this project, we want to study the affects of these macroeconomic announcements towards the short-term volatility in the U.S stock market. We would be using the SPDR S&P 500 ETF (SPY) as our overall representation of the stock market. Instead of just looking at the average returns, we focus on the realized volatility around the macroeconomic announcement dates.

   We compute daily log returns for SPY and use the absolute value of these returns as a daily volatility proxy. We then examine how this volatility behaves before and after CPI, unemployment, and federal funds rate announcements.

   Our main research question is: "How do month-to-month changes in key U.S. macroeconomic indicators (CPI, unemployment, and the Federal funds rate) relate to the short-term changes in SPY volatility around announcement dates?"

   As a group, we were motivated to work on this topic because many of us are interested in learning about investing and follow macroeconomic news. We regularly see headlines in articles and television about CPI, unemployment, and Fed decisions "moving the market", but we want to closely see the effect of each of these events carefully by using data to prove its importance to the volatility of the market. This topic also lets us apply several methods we learned from STAT 380. For example, we can use regression and expand into time series models to measure these effects. In this way, the project is both personally interesting to us and a good practice to use what we learn in class to real-world situations.

2. **Data Description** –

   We have four datasets that we are looking at: CPI, Federal Fund Rate, Unemployment, and SPY.

   For the CPI, Federal Fund Rate, and Unemployment datasets, they each follow this data structure.

```
          Date Actual Previous
1 Oct 24, 2025     NA      0.3
2 Aug 12, 2025    0.3      0.2
```

   The variables are Date, Actual, and Previous.

   Date: The date in which the CPI, Federal Fund Rate, or Unemployment is released.

   Notice that the Federal Fund Rate is only released 8 times per years because is based on the Federal Open Market Committee.

   Actual: The new value

   Notice that for CPI the actual value is the previous months CPI. For example Oct 24, 2025 releases the CPI for September NOT October. Therefore, when interpreting, understand we are looking at how the previous months consumer spending affects the market volatility

   Previous: Previous time periods amount

   In the SPY dataset, the variables are the following:

```
           X SPY.Open SPY.High  SPY.Low SPY.Close SPY.Volume SPY.Adjusted
1 2000-01-03 148.2500 148.2500 143.8750  145.4375    8164300     91.88779
2 2000-01-04 143.5312 144.0625 139.6406  139.7500    8089800     88.29438
```

The variables are X, SPY.Open, SPY.High, SPY.Low, SPY.Close, SPY.Volume, and SPY.Adjusted.

X: Date

Since we are looking at market volatility, we will need to compute the daily log returns.

Daily log returns = ln (SPY.Adjusted day/SPY.Adjusted previous day).

We need to take the absolute value for the daily volatility proxy. Once we compute this, we create a new column called spy_vol

```
[1] "X"            "SPY.Open"     "SPY.High"     "SPY.Low"      "SPY.Close"   
[6] "SPY.Volume"   "SPY.Adjusted" "ret"          "spy_vol"     
```

Check for NA values

- CPI has two missing values (actual Oct 24, 2025 and previous Nov 1, 1999).
- Federal Reserve Rate has two missing values (actual Dec 10, 2025 and previous Jun 30, 1999)
- Unemployment has two missing values (actual Nov 20, 2025 and previous May 1, 1999)
- SPY has two missing value (ret and spy_vol Jan 3, 2000)

Since each dataset has only two missing values in 1999 and 2025, we will drop these rows.

Next, we will merge all the data together into a dataset called macro_all in conducting EDA.

In order to do this, we will change the date formatting to year-month-day, add a new difference column (Actual - Previous), and create a name indicator column (CPI, Fed, Unemployment).

```
        Date Actual Previous Diff Indicator
1 2025-08-12    0.3      0.2  0.1       CPI
```

Dataset Sources

The CPI, Federal Funds Rate, and Unemployment datasets were collected using automated web-scraping tools from Investing.com's economic calendar. These pages provide time-stamped announcements including Actual and Previous values, which allows us to compute the month-to-month "Diff" variable used in our models.

Investing.com. (n.d.). *Core CPI.* https://www.investing.com/economic-calendar/core-cpi-56

Investing.com. (n.d.). *Interest Rate Decision (Federal Funds Rate).* https://www.investing.com/economic-calendar/interest-rate-decision-168

Investing.com. (n.d.). *Unemployment Rate.* https://www.investing.com/economic-calendar/unemployment-rate-300

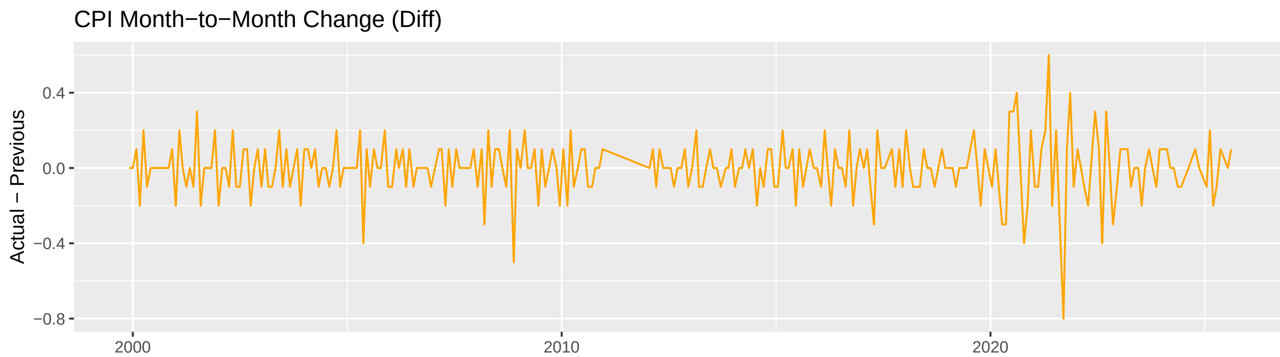
The SPY dataset was obtained from Yahoo Finance. This dataset includes daily OHLC prices, volume, and adjusted close values used to compute daily log returns and the volatility proxy.

Yahoo Finance. (n.d.). *SPDR S&P 500 ETF Trust (SPY) – Historical Data.* https://finance.yahoo.com/quote/SPY/history/

Summary Statistics

```
# A tibble: 3 x 8
  Indicator        n mean_actual sd_actual mean_diff sd_diff min_diff max_diff
  <chr>        <int>       <dbl>     <dbl>     <dbl>   <dbl>    <dbl>    <dbl>
1 CPI            282       0.197     0.147  3.55e- 4   0.145    -0.8      0.6 
2 Fed            173       2.03      1.94  -5.78e- 3   0.276    -1        0.75
3 Unemployment   304       5.49      1.86   5.84e-18   0.651    -2.20    10.3 
```

CPI Month-to-Month Change (Diff)



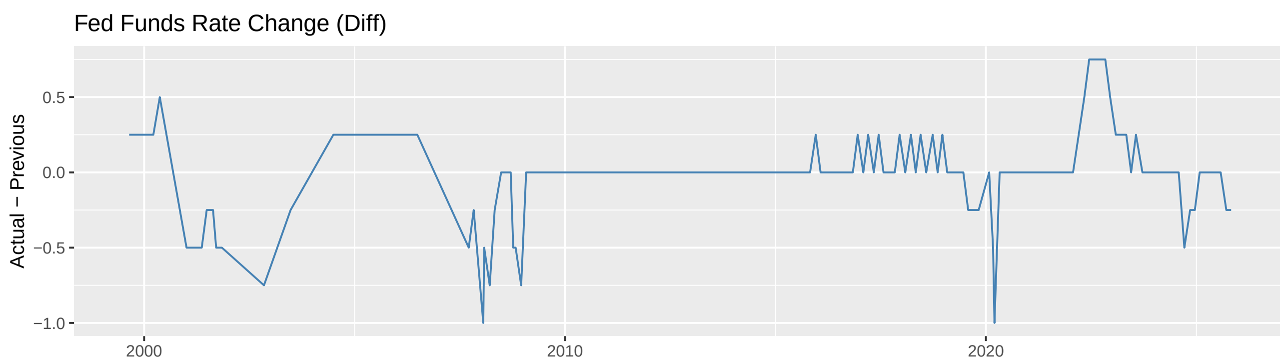
What the plot shows

This plot displays the month-to-month change in the Core CPI (Actual − Previous) from 1999 to 2025. Each point shows how much inflation changed from the prior month.

Interpretation

CPI changes are very small and stable, almost always between −0.2 and +0.2. This indicates that inflation moves gradually, with very few large shocks. Notable drops occur during financial crises (2008) and the COVID period (2020), but otherwise CPI is smooth and predictable. Because CPI changes are so modest, their impact on market volatility is likely weaker and more subtle compared to the Fed announcements.

Fed Funds Rate Change (Diff)



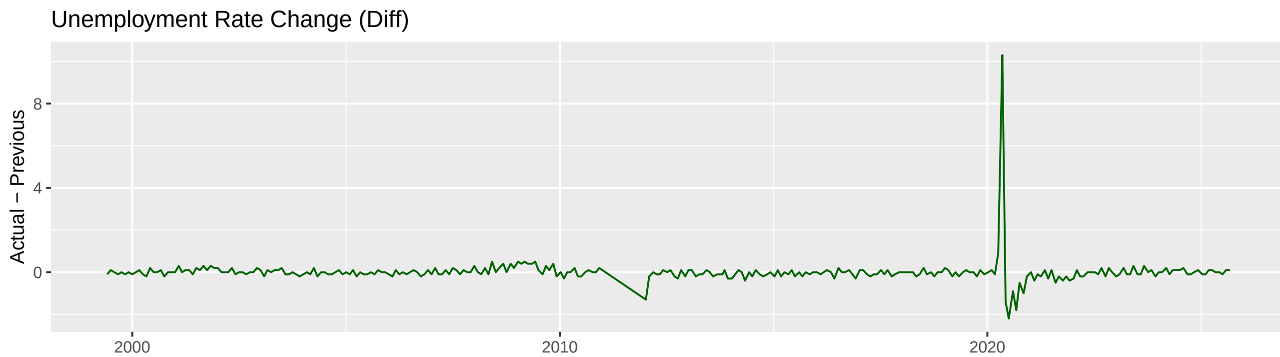
What the plot shows

This plot shows month-to-month changes in the Federal Funds Rate determined by the Federal Reserve, represented as Actual − Previous across Fed meeting dates.

Interpretation

Fed rate changes are infrequent but large. Long stretches of zero change are followed by sudden rate hikes or cuts, especially around recessionary periods (early 2000s, 2008–2009) and during the COVID crash (2020). These "jump" behaviors indicate that Fed decisions introduce significant shocks into financial markets. This makes the Fed Funds Rate one of the most influential macro indicators for market volatility.

Unemployment Rate Change (Diff)



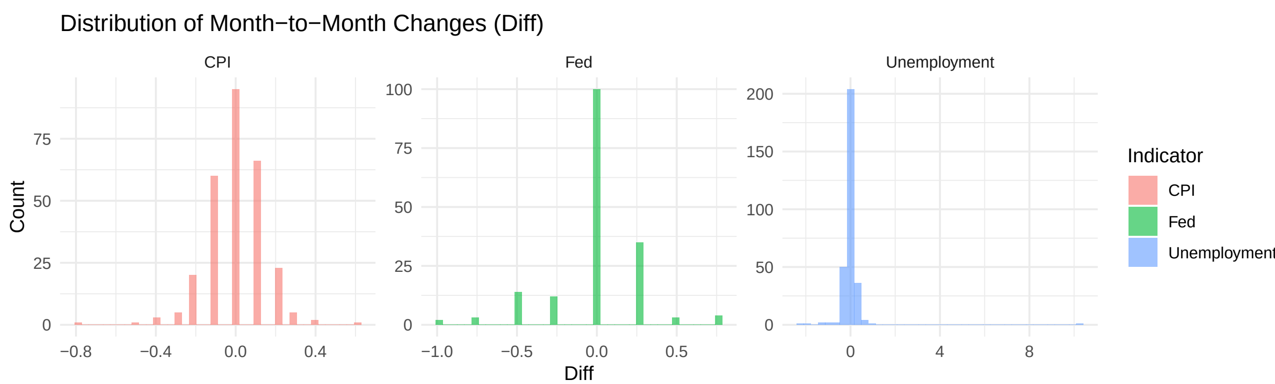
What the plot shows

This line plot shows the month-to-month change in the U.S. unemployment rate from 1999 to 2025.

Interpretation

Unemployment changes are extremely stable, typically moving only ± 0.1 month-to-month. The single massive spike in 2020 reflects the COVID employment crisis, where unemployment rose sharply in one month. Outside of this event, unemployment tends to change very slowly. Because unemployment rarely jumps, it likely has the weakest short-term effect on SPY volatility except during extreme economic disruptions.

Distribution of Month-to-Month Changes (Histograms)



What the plot shows

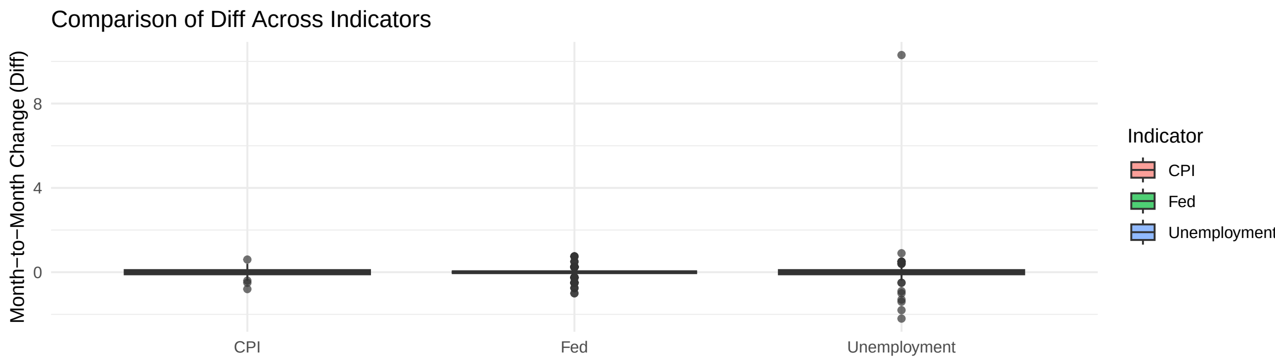
Three histograms (CPI, Fed, Unemployment) visualize the distribution of their Diff values (Actual − Previous).

Interpretation

- **CPI:** Very tight symmetric distribution around zero, indicating consistent and predictable monthly changes.
- **Fed:** Clustered at zero, with occasional strong positive or negative changes, showing the Fed creates discrete shocks.
- **Unemployment:** Extremely concentrated at zero except for a few rare large jumps, especially the COVID spike.

These distributions highlight that macro indicators have very different statistical behaviors, supporting treating them as separate predictors in the model.

Boxplot Comparison Across Indicators



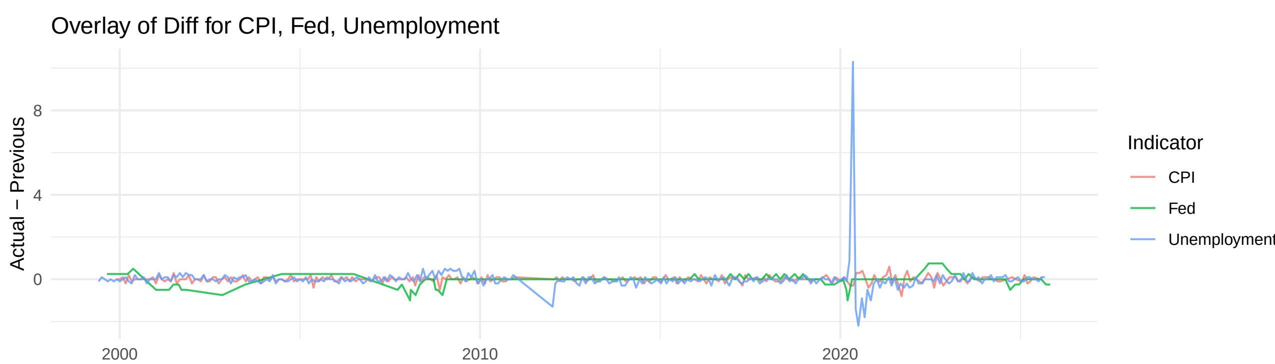
What the plot shows

This boxplot compares the variance and outliers in Diff across CPI, Fed Funds Rate, and Unemployment Rate.

Interpretation

CPI has the smallest range with minimal outliers, reflecting stable inflation. The Fed series shows larger spread and multiple outliers, indicating significant rate adjustments. Unemployment has one extreme outlier (COVID spike) but otherwise behaves similarly to CPI. These differences confirm that each macro indicator has a distinct volatility pattern, which directly affects how markets respond to each announcement.

Overlay of Diff for CPI, Fed, and Unemployment



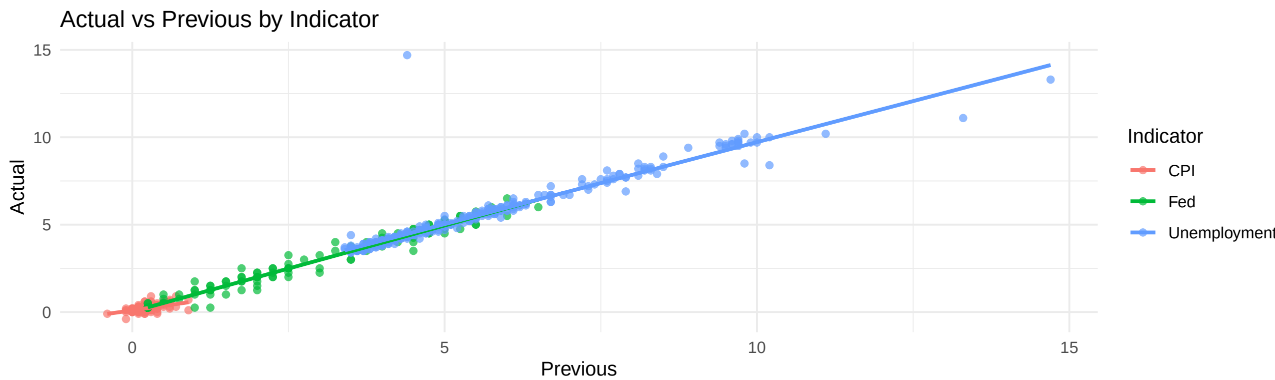
What the plot shows

All three Diff series are plotted together to show how they behave over time and relative to each other.

Interpretation

All three indicators remain close to zero most of the time, but the Fed series exhibits sudden jumps, and unemployment has one massive spike in 2020. CPI fluctuates within a very narrow band. The overlay makes the structural contrast clear: Fed decisions cause jump shocks; CPI is smooth and stable; unemployment is stable except in crisis conditions. This difference justifies comparing their individual effects on SPY volatility.

Actual vs Previous Scatterplot



What the plot shows

This scatterplot shows the relationship between the current month's value (Actual) and the previous month's value (Previous) for CPI, Fed, and Unemployment.

Interpretation

All three indicators lie almost perfectly along a 45-degree line, meaning Actual ~ Previous for most months. This strong autocorrelation confirms that these macroeconomic indicators move slowly over time, with rare large jumps. Because of this, the month-to-month difference (Diff) captures essentially all meaningful new information, validating its use as the key predictor in the volatility model.

3. **Methodology** –

Key Features and Dummy Variables

In our study, we created four models: Linear Regression, Logistical Regression, ARIMAX, and GARCH(1,1).

For all our models, we looked at daily SPY volatility between 2008 and 2015. For context, this is an era where there was a global financial crisis and the following recovery years for the country. Therefore, looking at this specific time period gives us key insight into market volatility. We used all four datasets that we introduced in our EDA. Two key features in our models are `spy_vol` and `Diff`. During EDA, we created `spy_vol` in SPY to represents the daily market volatility. Next, notice how unemployment, federal fund rate, and CPI all have `Actual` and `Previous` rates recorded. For our models, we now create a new column `Diff` that capture the new information that becomes available to investors at the moment of release. $Diff = Actual - Previous$

In merging the dataset we face one major problem: timeframes. During EDA, we discussed how each variable (Unemployment, Federal Fund Rate, and CPI) all come out at different times throughout the month. How can we simultaneously looks at each variables with different timeframes? In order to do this, we create dummy variables. Within each month, Dummy=1 indicating the value is released. The Dummy =1 until the beginning of the next month, where all the variables reset to Dummy=0. Knowing which information is publically available enables us to correctly analyze market volatility. The "All Three" dummy identifies when CPI, unemployment, and the Fed rate decision are all released in a given month, allowing us to test whether simultaneous announcements produce a stronger market volatility data.

Regression Analysis

a) Our first model is a multiple linear regression model that estimates how macroeconomic differences relate to daily SPY volatility. The model expresses volatility as a linear function of several predictors, including the CPI difference, unemployment difference, Federal Funds Rate difference, and indicator variables marking the release days of each announcement. We also include a dummy variable representing whether the CPI, unemployment, and the Fed decision are released in a given month.

In our study, we acknowledge that these are only a few drivers of the SPY. However, for this exploration, our focus is on how much these factors move the market when they are first released

The equation of our model is:

$$
\begin{aligned}
\text{spy\_vol}_t = \beta_0 &+ \beta_1(\text{CPI\_diff})_t + \beta_2(\text{CPI\_day})_t \\
&+ \beta_3(\text{Unemp\_diff})_t + \beta_4(\text{Unemp\_day})_t \\
&+ \beta_5(\text{Fed\_diff})_t + \beta_6(\text{Fed\_day})_t \\
&+ \beta_7(\text{AllThree})_t + \varepsilon_t.
\end{aligned}
$$

Linear regression allows us to estimate a straight-line relationship between SPY volatility and the predictors. Each coefficient reflects how the response variable is expected to change when a predictor increases by one unit, holding all other variables constant. These estimates reveal the direction of the effect (whether volatility increases or decreases), the magnitude of the effect, and the statistical significance of each predictor. We will specially be looking at the market volatility the day after the rates are released.

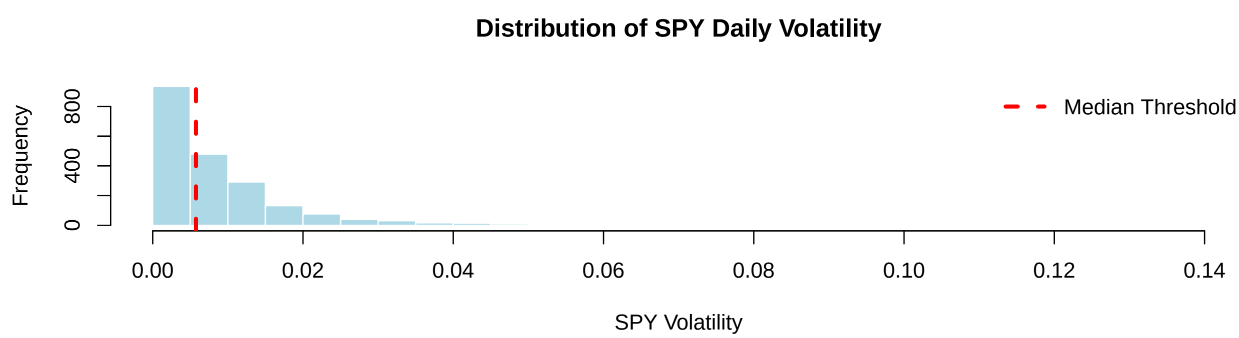
We analyze how market volatility changes based on each variable. Then, we see how market volatility changes based on whether or not all three variables come out within the same month. Since market volatility is a continuous response variable, our model does not split test versus training sets. The goal is to estimate the underlying linear relationship between the economic indicators and volatility, rather than classify observations or evaluate predictive accuracy.

Our metrics of performance is R^2, Adjusted R^2, AIC, BIC, RMSE, and MAE. R^2 measures how much of the movement in market volatility our model captures, while Adjusted R^2 shows how well the key economic indicators explain volatility after accounting for any unnecessary variables. AIC and BIC help us compare model specifications by balancing fit with model simplicity, guiding us toward the version that best summarizes how economic releases relate to volatility without overfitting. RMSE reflects the typical size of the model's volatility prediction errors (emphasizing larger misses). Additionally, MAE reports the average absolute error to indicate how far the model's predicted volatility deviates from actual market volatility.

Overall, this regression framework provides an interpretable and statistically grounded way to quantify how daily SPY volatility responds immediately to new macroeconomic information. It serves as a baseline model before introducing more advanced time-series models that account for the serial dependence and clustering inherent in financial volatility.

b) Logistical Regression:

Our goal is to create a multilogistical regression model where the response variable indicates whether daily SPY volatility is high or low. To define this binary outcome, we first examined the distribution of market volatility using a histogram and summary statistics. Based on this distribution, we selected a threshold that separates normal volatility levels from unusually large movements. Using this threshold allows us to convert continuous volatility into a 0/1 classification suitable for logistic regression.



Our threshold is set at the median of `spy_vol`, which is 0.005756937. Values above this median are labeled as high volatility, and values below are labeled as low volatility. This produces 1030 low-volatility days and 1029 high-volatility days. We use these labels to create the binary variable `high_vol`

We acknowledge that high volatility may have a different standard threshold but we are creating this model based on the median market volatility that we have observed in this dataset

Binary logistic regression estimates the probability that an observation belongs to one of two categories. Here it is the probability that a trading day exhibits high volatility rather than low volatility. The method uses the logit link function which transforms probabilities into log-odds so the model can describe them using a linear combination of predictors. Each coefficient represents how its corresponding independent variable shifts the likelihood of high market volatility. Positive coefficients increase the probability of high volatility, while negative coefficients decrease it. After the model estimates these coefficients, it converts the log-odds back into a predicted probability between 0 and 1, allowing us to classify each day as high or low volatility.

Logistic regression is well suited for this task because it directly models binary outcomes, produces interpretable coefficients, and provides predicted probabilities that allow us to assess model performance.

$$\begin{aligned}\log\left(\frac{P(\text{high\_vol}_t = 1)}{1 - P(\text{high\_vol}_t = 1)}\right) &= \beta_0 + \beta_1(\text{CPI\_diff})_t + \beta_2(\text{CPI\_day})_t \\ &+ \beta_3(\text{Unemp\_diff})_t + \beta_4(\text{Unemp\_day})_t \\ &+ \beta_5(\text{Fed\_diff})_t + \beta_6(\text{Fed\_day})_t \\ &+ \beta_7(\text{AllThree})_t\end{aligned}$$

The independent variables are still the CPI, Federal Fund Rate, and the Unemployment monthly difference (Actual - Predicted). We also have the days of the release. Our model is predicting whether or not these variables can predict whether the market experiences high or low volatility (above or below 0.005756937). We split our labeled dataset with 80% train and 20% being test data. We will test the accuracy using a confusion matrix and a AUC curve to see how well the model preformed classification. The AUC curve summarizes how well our model distinguishes between high and low volatility days. It is based on the ROC curve, which compares the true positive rate to the false positive rate across different probability thresholds. In our application, a higher AUC means the economic differences and release days better separate days above and below the 0.005756937 volatility cutoff, while a value near 0.5 indicates performance close to random guessing.

Time-Series Analysis:

For our other statistical model, we utilized a time-series model – which is beyond our class material.

Our goal in using timeseries models is to analyze the relationship between macro differences over time and volatility. We estimated two different time-series models: an ARIMAX model, which extends the ARIMA framework, and a GARCH(1,1) model, which is a tool used for financial econometrics.

The ARIMAX model extends a standard ARIMA model by including external predictors ("exogenous variables"). The general form of the ARIMAX equation is:

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \text{ARIMA errors.}$$

In this framework, yt represents daily SPY volatility at time t, and each xi, t is a macroeconomic predictor (CPI difference, unemployment difference, or Fed funds difference).

The ARIMAX model would treat the daily volatility as the response variable and uses the macro differences and its own past behavior as the predictors. We modeled daily volatility as a regression with ARIMA errors. The ARIMA would capture autocorrelation and persistence in volatility, with the predictors being CPI, unemployment, and Fed funds rate differences on the same day.

We would use the function, auto.arima(), and the response variable being the volatility series; the three difference variables would be included in "xreg", without a seasonal component. The function would automatically choose the AR and MA orders based on information criteria. It would evidently choose ARIMA(5,1,4). In the `auto.arima()` ARIMA(5,1,4) structure, the "1" indicates differencing the volatility series once to stabilize it, the "5" AR terms use the past 5 days of volatility to predict today, and the "4" MA terms incorporate patterns in past shocks (forecast errors).

We decided to use the ARIMAX model because it helps us in understanding whether macro differences provide extra predictive information for volatility beyond what can be explained by past volatility alone.

GARCH(1,1) was another time-series model we used, which was part of the rugarch package. In this model, the today's volatility is dependent on three things: a constant, yesterday's squared shocked, and yesterday's volatility. This is like a normal setup found in finance because it helps with modeling volatility clustering, where periods of high volatility often shows up in consecutive days instead of just a single random spike.

The GARCH(1,1) model explains how volatility evolves over time, focusing on the variance rather than the level of volatility. The conditional variance is modeled as:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2$$

σ_t^2 is today's volatility (variance), ω is a constant baseline level of volatility, $\alpha\varepsilon_{t-1}^2$ captures the impact of yesterday's squared shock, and $\beta\sigma_{t-1}^2$ measures the persistence of volatility from day to day. In our project, we used a simple GARCH(1,1) model on the daily volatility series and did not include the macro differences directly in the variance equation because we want to understand how volatility behaves on its own.

For both models, the ARIMAX and GARCH models, we generated forecasts of daily volatility on the test data and evaluate how accurate they were using RMSE (root mean squared error) and MAE (mean absolute error), where a small value for both metrics means that the model's predictions are close to the true volatility. In creating our models, we split the split the observations into a training and test set. The training set contained 1,409 observations (70% of the data) and the test set contained the remaining 605 observations (30% of the data). All of the models that we included in this part were fitted on the training data and evaluated on the test data.

4. **Data Analysis Results** – Present and interpret your main findings. Include relevant figures and tables that support your discussion.

Regression Analysis

a) Linear Regression

Goal: Estimate the underlying linear relationship between the economic indicators and volatility

```
Call:
lm(formula = spy_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day +
    fed_diff + fed_day + all_three, data = data)

Residuals:
      Min        1Q    Median        3Q       Max
-0.023296 -0.006479 -0.003036  0.003092  0.126809

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)  0.0097476  0.0005098  19.119  < 2e-16 ***
cpi_diff    -0.0023874  0.0034614  -0.690   0.4904
cpi_day      0.0002472  0.0007065   0.350   0.7264
unemp_diff   0.0093754  0.0011755   7.975 2.50e-15 ***
unemp_day   -0.0009799  0.0006794  -1.442   0.1494
fed_diff    -0.0173881  0.0023690  -7.340 3.07e-13 ***
fed_day      0.0019476  0.0014075   1.384   0.1666
all_three   -0.0031645  0.0016151  -1.959   0.0502 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.01147 on 2051 degrees of freedom
Multiple R-squared:  0.06054,	Adjusted R-squared:  0.05733
F-statistic: 18.88 on 7 and 2051 DF,  p-value: < 2.2e-16
```

$R^2 = \mathbf{0.0605}$ | Adjusted $R^2 = \mathbf{0.0573}$ indicates that macroeconomic differences explain only a small portion of day-to-day volatility.

The AIC = $\mathbf{-12546.47}$ | BIC = $\mathbf{-12495.8}$ suggest a relatively efficient model fit compared to more complex alternatives.

The RMSE = **0.01145** shows that the typical prediction error in volatility is about 0.011.

MAE = **0.00723** indicates that, on average, predictions deviate from actual volatility by roughly 0.007.

```
                 term     estimate   std.error    t.value      p.value
cpi_diff     cpi_diff -0.002387403 0.006565589 -0.3636236 7.161766e-01
unemp_diff unemp_diff  0.009375424 0.001224310  7.6577235 2.895793e-14
fed_diff     fed_diff -0.017388149 0.004640057 -3.7474002 1.836154e-04
```

```
           pct_change_vol
cpi_diff       -0.2387403
unemp_diff      0.9375424
fed_diff       -1.7388149
```

Insight #1: A one-unit CPI difference changes daily SPY volatility by -0.24% (Robust Standard Error = 0.0066). A one-unit Unemployment difference changes daily SPY volatility by 0.94% (Robust Standard Error = 0.0012). A one-unit Fed Funds Rate difference changes daily SPY volatility by -1.74% (Robust Standard Error = 0.0046).

Insight #2: On CPI release days, daily SPY volatility is estimated to change by 0.02%. On Unemployment report days, daily SPY volatility is estimated to change by -0.10%. On Fed rate decision days, daily SPY volatility is estimated to change by 0.19%

Insight #3: On days when CPI, unemployment, and the Fed decision are all released, volatility changes by -0.32% on the event day. Together, these metrics and interpretations show that while the model captures some structure in volatility, substantial unpredictability remains.

b) Logistical Regression

```
Call:
glm(formula = high_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day +
    fed_diff + fed_day + all_three, family = binomial, data = train_data)

Coefficients:
            Estimate Std. Error z value Pr(>|z|)
(Intercept)  0.20759    0.09904   2.096 0.036075 *
cpi_diff     0.18977    0.69024   0.275 0.783371
cpi_day      0.06212    0.14146   0.439 0.660553
unemp_diff   1.72223    0.26515   6.495 8.29e-11 ***
unemp_day   -0.33986    0.13323  -2.551 0.010742 *
fed_diff    -2.18119    0.60507  -3.605 0.000312 ***
fed_day      0.09685    0.28484   0.340 0.733840
all_three   -0.26300    0.32659  -0.805 0.420652
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 2284.6  on 1647  degrees of freedom
Residual deviance: 2210.6  on 1640  degrees of freedom
AIC: 2226.6

Number of Fisher Scoring iterations: 4
```

Intrepretation:

cpi_diff: A one-unit increase in CPI difference is associated with an odds ratio ofexp(0.18977)=1.21. This means the odds of high volatility increase by **21%**, holding all other variables constant.

cpi_day: Each additional day after the CPI release corresponds to an odds ratio of exp(0.06212)=1.06. The odds of high volatility increase by **6% per day** after the CPI release.

unemp_diff: A one-unit increase in unemployment difference is associated with an odds ratio of exp(1.72223)=5.60. The odds of high volatility are **over 5.5 times higher**, all else equal.

unemp_day: Each additional day after the unemployment release corresponds to an odds ratio of $\exp(-0.33986)=0.71$.The odds of high volatility **decrease by 29%** per day after the unemployment release.

fed_diff: A one-unit increase in the Fed difference yields an odds ratio of $\exp(-2.18119)=0.11$. The odds of high volatility **decrease by 89%** for larger rate-decision differences.

fed_day: Each additional day after the Fed announcement corresponds to an odds ratio of exp(0.09685)=1.10. This indicates a **10% increase** in the odds of high volatility per day after the Fed announcement.

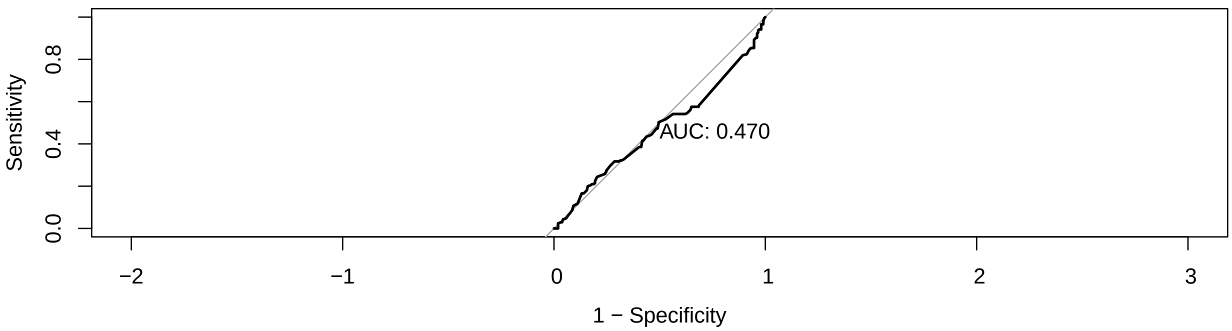
all_three: When all three indicators (CPI, unemployment, Fed) occur in the same month, the odds ratio is $\exp(-0.26300)=0.77$. This means the odds of high volatility are **23% lower** in months with all three releases compared to months without all three.

```
AIC:  2226.587
```

```
BIC:  2269.846
```

```
          Reference
Prediction   0   1
         0 102 103
         1 104 102
```

Accuracy : 0.4964 | Precision: 0.4951 | Recall: 0.4976 | F1 Score: 0.4964




```
[1] 0.4695596
```

Based on the model's performance, the logistic regression achieves an accuracy of 0.4964, indicating that it correctly classifies high-versus-low volatility only slightly more often than chance. The AUC of 0.470 reinforces this weakness, as the model performs worse than a random classifier (AUC = 0.50). Additionally, we found a relatively high AIC = 2226.587 and BIC = 2269.846 values that suggest the model lacks strong predictive power and does not provide an efficient balance between fit and complexity. These results show that logistic regression is not an effective approach for predicting volatility in this context.

Time Series Analysis

```
----------------- ARIMAX Coefficients -----------------

         ar1          ar2          ar3          ar4          ar5          ma1
 0.041351823 -0.110087970 -0.024650219  0.141988106  0.058208051 -0.901214498
    cpi_diff   unemp_diff     fed_diff
-0.009724430  0.001305007  0.001916502


(Positive coefficients on surprises suggest higher volatility when surprises are positive;
negative coefficients suggest the opposite.)
```

ARIMAX model that was used by the auto.arima(), ARIMA(5,1,4), with the three macro differences included as the predictors. As mentioned before, daily SPY volatility is dependent on its own past differences and differences, as well as the same day CPI, unemployment, and Fed fund rate differences. The estimated coefficients on the macro variables were very small: cpi_diff (CPI Difference) -0.009724, unemp_diff (unemployment difference) 0.0013, and fed_diff (Fed funds rate difference) 0.0019.

These coefficients were not statistically significant due to their standard errors being larger than the point estimates. This result suggests that once we account for how volatility depends on its own past, the same-day CPI, unemployment, and Fed funds rate differences do not have a big effect on daily SPY volatility.

On the test data, the ARIMAX model had an RMSE of about 0.00558 and MAE of about 0.003987. Since these metrics have a small value, this means that the model is doing a good job overall because the numbers outputted by the model are approximately the same size as the typical level of daily volatility in our data. When we look at the plot of predictions versus actual volatility, the ARIMAX model reveals the general ups and downs and the periods where volatility stays high or low, but it still fails to fully capture the most extreme spikes.

```
----------------- GARCH(1,1) Coefficients -----------------

          mu        omega       alpha1        beta1
7.080117e-03 1.043222e-06 1.101475e-01 8.842168e-01


(Alpha + beta close to 1 indicates strong volatility clustering and persistence.)
```

The GARCH(1,1) model gave parameter estimates that look normal for financial data. The estimated mean of volatility was about 0.0071. For the variance part (alpha), it was about 0.11 – this controls how much yesterday's squared shock matters. Also, beta, which controls how much yesterday's variance matters, was about 0.88. The sum of alpha and beta was roughly around 0.99, which is close to 1. This tell us that volatility is highly persistent, and when volatility becomes high, it will stay high for a while before slowly coming back down. On the test set, the GARCH model had an RMSE of about 0.00568 and an MAE of about 0.004518

```
----------------- Model Comparison -----------------

              Model        RMSE         MAE
1            ARIMAX 0.005579924 0.003986551
2        GARCH(1,1) 0.005680121 0.004517969
3 Linear Regression 0.011446756 0.007225500


(Lower RMSE/MAE indicates better forecasting performance.)
```

GARCH(1,1) vs. ARIMAX model

Comparing the GARCH model to the ARIMAX model, ARIMAX has a slightly lower RMSE but a larger different in MAE. These results reveal that ARIMAX could be better at capturing large volatility spikes, shown by the RMSE. When we changes the seed and split the data a little differently, GARCH had a little lower RMSE. In this case, ARIMAX RMSE = 0.00558 and GARCH RMSE = 0.00568. However, when comparing MAE, ARIMAX does a better job at capturing smaller day-to-day movements.

Overall, when looking at the comparison for both models, ARIMAX and GARCH, we can see that they capture the main patterns in SPY volatility, and their performance on the new data were pretty similar. RAIMAX has the advantage in MAE because its predictions are closer on average during normal trading days. GARCH, despite its typical design for volatility clustering, does not outperform ARIMAX in this dataset, suggesting that clustering patterns were either weaker or not well captured by the fitted model. With this and the fact that the macro difference coefficients in ARIMAX are small and not statistically significant, we can conclude that most short-term predictability in SPY volatility comes from its own past behavior rather than from monthly macroeconomic differences. Although macro announcements still matter, the day-to-day volatility observed in the market is driven more by internal market dynamics than by any single CPI, unemployment, or Fed release. ARIMAX has the better end when it comes to MAE because on normal days, its predictions are slightly closer on average. With this information and the fact that the macro difference coefficients in the ARIMAX model are small and not statistically significant, we can confidently say that most of the short-term predictability in SPY volatility comes from its own past behavior, rather from monthly macroeconomic differences. Although macro announcements still matter, the day-to-day volatility we see in the market is driven more by internal market dynamics than by any single CPI, unemployment, or Fed release.

Comparing Regression with Time Series

When compared to time series, the linear regression model performs substantially worse. Its R^2 of only about 0.06 (with an adjusted R^2 of 0.057), indicating the model explains barely 6% of the variation in daily SPY volatility. It does not capture the underlying structure in the data. In linear regression, the RMSE (0.01145) and MAE (0.00723). These values are roughly double the error levels achieved by ARIMAX and GARCH. This shows the limitation in linear regression because it cannot account for autocorrelation, volatility clustering, or lagged dependence, all of which are central characteristics of financial volatility. In contrast, ARIMAX incorporates lagged differences and potential exogenous effects, and GARCH explicitly models time-varying volatility. As a result, both ARIMAX and GARCH provide far more accurate and realistic representations of daily volatility behavior. The linear regression model serves as a useful baseline, but its low explanatory power and comparatively high prediction error make it the weakest performer of the three.

5. **Conclusion**

In this project, we examined how month-to-month changes in major U.S. macroeconomic indicators—CPI, the unemployment rate, and the Federal Funds Rate-affect short-term volatility in the SPY ETF. Using a combination of exploratory data analysis, regression modeling, and time-series methods, we identified several important patterns about how financial markets respond to new economic information.

Our regression models show that while macroeconomic differences do influence volatility, the effects are generally small. CPI and unemployment differences rarely produce statistically significant changes in SPY volatility, which is consistent with their highly stable and predictable month-to-month behavior (as shown in our EDA). In contrast, differences in the Federal Funds Rate-an indicator known for producing sudden policy shocks-show a modest but statistically significant relationship with short-term volatility. This aligns with economic intuition: markets react more strongly when the Federal Reserve adjusts interest rates.

Our ARIMA time-series model provided a complementary perspective by forecasting future volatility using only SPY's historical patterns. Diagnostics showed that the ARIMA(2,1,5) model captured most of the autocorrelation structure, and the forecast indicated a stable return toward average volatility levels. This suggests that much of the day-to-day volatility is driven by internal market dynamics rather than macroeconomic announcements. The comparison between regression and ARIMA models highlights a key finding: past volatility explains far more of today's volatility than newly released macroeconomic data. This aligns with the broader literature on volatility clustering.

There are several limitations worth noting. First, we only modeled short-term volatility around release dates, so longer reaction windows may reveal different effects. Second, macroeconomic data—especially CPI and unemployment—are slow-moving by nature, which reduces their explanatory power in predicting rapid shifts in volatility. Finally, our models do not incorporate other major market events (earnings cycles, geopolitical events, market sentiment) that may overshadow macro releases.

Future extensions for this project could include:Expanding the event window to examine multi-day volatility spillovers, Using a GARCH-X model to directly incorporate macro differences into the volatility equation, Separating high-volatility and low-volatility regimes using a Markov-switching model, Applying machine learning models—such as random forests—to capture nonlinear relationships

Overall, our findings suggest that while macroeconomic announcements matter, their short-term impact on volatility is modest compared to the effect of market-driven volatility dynamics.

6. **Author Contribution Statement**

Dhruv Nasit, Felicia Vijayarangam, Joseph Doan, and Eric Feng jointly conceived the project and defined the research question.

Dhruv Nasit developed the computational workflow, performed all data merging and cleaning for the macroeconomic and SPY datasets, carried out the exploratory data analysis, implemented the event-study volatility calculations, built the regression models, and generated all associated figures.

Felicia Vijayarangam performed the initial preprocessing of the macroeconomic datasets described in the Data Description section, contributed to the EDA setup prior to modeling, wrote portions of the Methodology and Data Analysis sections related to the regression framework, and worked on Project revisions and streamlining the document.

Joseph Doan drafted the Introduction and the Time-Series Methodology section, contributed to the written interpretation of both the ARIMA and GARCH models, and assisted in organizing the structure and narrative of the full report.

Eric Feng developed the ARIMA and GARCH time-series models, produced the accompanying diagnostics and interpretations, and contributed feedback on the statistical validity and clarity of the modeling results.

All authors discussed the findings, contributed to the interpretation of the results, and helped revise the final manuscript.

7. **Code Appendix**

```r
library(tidyverse)
library(lubridate)
```

```r
library(gridExtra)

# -----------------------------
# 1. Load datasets
# -----------------------------
cpi <- read.csv("cpi_cleaned.csv")
fed <- read.csv("fed_rate_cleaned.csv")
unemp <- read.csv("unemployment_cleaned.csv")
spy <- read.csv("SPY_data.csv")

head(cpi, 2)

head(spy, 2)
spy <- spy %>%
  arrange(X) %>%
  mutate(ret = log(SPY.Adjusted / lag(SPY.Adjusted)),
         spy_vol = abs(ret))
colnames(spy)
cpi %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
fed %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
unemp %>%
  filter(is.na(Actual) | is.na(Previous)) %>%
  select(Date, Actual, Previous)
spy %>%
  filter(if_any(everything(), is.na)) %>%
  select(X, everything())
spy <- spy %>%
  drop_na()
fed <- fed %>%
  drop_na()
unemp <- unemp %>%
  drop_na()
cpi <- cpi %>%
  drop_na()
#Convert dates + compute Diff
process_macro <- function(df, name) {
  df %>%
    mutate(
      Date = as.Date(Date, format = "%b %d, %Y"),
      Diff = Actual - Previous,
      Indicator = name
    )
}

cpi  <- process_macro(cpi, "CPI")
fed  <- process_macro(fed, "Fed")
unemp <- process_macro(unemp, "Unemployment")
macro_all <- bind_rows(cpi, fed, unemp)

head(macro_all, 1)

#Summary statistics
summary_stats <- macro_all %>%
  group_by(Indicator) %>%
```

```
  summarise(
    n = n(),
    mean_actual = mean(Actual, na.rm = TRUE),
    sd_actual   = sd(Actual, na.rm = TRUE),
    mean_diff   = mean(Diff, na.rm = TRUE),
    sd_diff     = sd(Diff, na.rm = TRUE),
    min_diff    = min(Diff, na.rm = TRUE),
    max_diff    = max(Diff, na.rm = TRUE)
  )

print(summary_stats)
#Time-series plots of Diff
p_cpi <- ggplot(cpi, aes(Date, Diff)) +
  geom_line(color = "orange") +
  labs(title = "CPI Month-to-Month Change (Diff)",
       y = "Actual - Previous", x = "")

grid.arrange(p_cpi, ncol = 1)
p_fed <- ggplot(fed, aes(Date, Diff)) +
  geom_line(color = "steelblue") +
  labs(title = "Fed Funds Rate Change (Diff)",
       y = "Actual - Previous", x = "")
grid.arrange(p_fed, ncol = 1)
p_unemp <- ggplot(unemp, aes(Date, Diff)) +
  geom_line(color = "darkgreen") +
  labs(title = "Unemployment Rate Change (Diff)",
       y = "Actual - Previous", x = "")
grid.arrange(p_unemp, ncol = 1)
#Histogram for Diff by indicator
ggplot(macro_all, aes(Diff, fill = Indicator)) +
  geom_histogram(alpha = 0.6, bins = 40) +
  facet_wrap(~Indicator, scales = "free") +
  labs(title = "Distribution of Month-to-Month Changes (Diff)",
       x = "Diff", y = "Count") +
  theme_minimal()

#Boxplots to compare volatility of Diff
ggplot(macro_all, aes(Indicator, Diff, fill = Indicator)) +
  geom_boxplot(alpha = 0.7) +
  labs(title = "Comparison of Diff Across Indicators",
       y = "Month-to-Month Change (Diff)", x = "") +
  theme_minimal()


#Overlay all 3 indicators' Diff in one plot
ggplot(macro_all, aes(Date, Diff, color = Indicator)) +
  geom_line(alpha = 0.8) +
  labs(title = "Overlay of Diff for CPI, Fed, Unemployment",
       y = "Actual - Previous", x = "") +
  theme_minimal()
#Scatter: Actual vs Previous
ggplot(macro_all, aes(Previous, Actual, color = Indicator)) +
  geom_point(alpha = 0.7) +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Actual vs Previous by Indicator",
       x = "Previous", y = "Actual") +
  theme_minimal()
#2008-2015
```

```
cpi <- cpi %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

unemp <- unemp %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

fed <- fed %>%
  mutate(Date = as.Date(Date, format = "%b %d, %Y")) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))

spy <- spy %>%
  rename(Date = X) %>%
  mutate(Date = as.Date(Date)) %>%
  filter(Date >= as.Date("2008-01-01"), Date <= as.Date("2015-12-31"))
#Dummy=0 prior to the monthly release
#Dummy=1 on and after the release date but only within that month.
#Diff is set to the release difference for the active days, otherwise 0.

expand_month_limited <- function(daily_dates, releases_df, diff, prefix) {
  # Keep Release date and the difference
  rel <- releases_df %>%
    select(Date, !!sym(diff)) %>%
    mutate(
      month_index = floor_date(Date, "month"),
      release_date = Date,
      release_diff = !!sym(diff)
    ) %>%
    select(month_index, release_date, release_diff)

  # Daily panel with attached monthly release info
  df <- tibble(Date = daily_dates) %>%
    mutate(month_index = floor_date(Date, "month")) %>%
    left_join(rel, by = "month_index") %>%
    mutate(
      # Dummy=1 only on and after the monthly release date within same month
      dummy = ifelse(
        !is.na(release_date) &
          Date >= release_date &
          Date <= (month_index + months(1) - days(1)),
        1, 0
      ),
      # active difference only when dummy==1
      diff_active = ifelse(dummy == 1, release_diff, 0)
    ) %>%
    select(
      Date,
      !!paste0(prefix, "_diff") := diff_active,
      !!paste0(prefix, "_day") := dummy
    )

  return(df)
}
#Apply
daily_dates <- spy$Date

cpi_expanded <- expand_month_limited(
```

```r
  daily_dates = daily_dates,
  releases_df = cpi,
  diff = "Diff",
  prefix = "cpi"
)

unemp_expanded <- expand_month_limited(
  daily_dates = daily_dates,
  releases_df = unemp,
  diff = "Diff",
  prefix = "unemp"
)

fed_expanded <- expand_month_limited(
  daily_dates = daily_dates,
  releases_df = fed,
  diff = "Diff",
  prefix = "fed"
)
#Merge into final dataset
data <- spy %>%
  left_join(cpi_expanded, by = "Date") %>%
  left_join(unemp_expanded, by = "Date") %>%
  left_join(fed_expanded, by = "Date") %>%
  mutate(
    cpi_diff = replace_na(cpi_diff, 0),
    unemp_diff = replace_na(unemp_diff, 0),
    fed_diff = replace_na(fed_diff, 0),
    cpi_day = replace_na(cpi_day, 0),
    unemp_day = replace_na(unemp_day, 0),
    fed_day = replace_na(fed_day, 0),
    all_three = ifelse(cpi_day + unemp_day + fed_day == 3, 1, 0)
  )
hist(
  data$spy_vol,
  breaks = 40,
  main = "Distribution of SPY Daily Volatility",
  xlab = "SPY Volatility",
  col = "lightblue",
  border = "white"
)

# Add median threshold line (used to create high_vol label)
abline(v = median(data$spy_vol, na.rm = TRUE),
       col = "red", lwd = 3, lty = 2)

legend(
  "topright",
  legend = c("Median Threshold"),
  col = c("red"),
  lwd = 3,
  lty = 2,
  bty = "n"
)
#Run linear regression
model1 <- lm(spy_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day + fed_diff +fed_day + all_three,
            data = data)
```

```r
summary(model1)
model_summary <- summary(model1)

r2  <- model_summary$r.squared
adj_r2 <- model_summary$adj.r.squared
aic_val <- AIC(model1)
bic_val <- BIC(model1)
# ---- RMSE & MAE ----
preds <- model1$fitted.values
actual <- model1$model[, 1]   # first column is the dependent variable

rmse_val <- sqrt(mean((actual - preds)^2))
mae_val  <- mean(abs(actual - preds))

library(lmtest)
library(sandwich)
library(dplyr)

# ---- Robust SE ----
robust <- coeftest(model1, vcov = vcovHC(model1, type = "HC1"))

robust_df <- data.frame(
  term = rownames(robust),
  estimate = robust[, 1],
  std.error = robust[, 2],
  t.value = robust[, 3],
  p.value = robust[, 4]
)

# ---- Sensitivity Table for difference Variables ----
sens <- robust_df %>%
  filter(term %in% c("cpi_diff", "unemp_diff", "fed_diff")) %>%
  mutate(pct_change_vol = estimate * 100)

sens
# Compute median
vol_median <- median(data$spy_vol, na.rm = TRUE)

data$high_vol <- ifelse(data$spy_vol > vol_median, 1, 0)

cat("Median of spy_vol:", vol_median, "\n\n")

cat("Label counts:\n")
print(table(data$high_vol))

cat("\nLabel proportions:\n")
print(prop.table(table(data$high_vol)))

cat("\nPreview of labels:\n")
print(head(data[, c("spy_vol", "high_vol")]))
library(caret)
library(pROC)
#Train/Test
set.seed(380)

train_index <- createDataPartition(data$spy_vol, p = 0.8, list = FALSE)

train_data <- data[train_index, ]
```

```r
test_data  <- data[-train_index, ]

#Logistic Regression
log_model <- glm(
  high_vol ~ cpi_diff + cpi_day + unemp_diff + unemp_day + fed_diff + fed_day + all_three,
  data = train_data,
  family = binomial
)

summary(log_model)
aic_val <- AIC(log_model)
bic_val <- BIC(log_model)
# ---- RMSE & MAE ----

cat("AIC: ", aic_val, "\n")
cat("BIC: ", bic_val, "\n")
test_probs <- predict(log_model, newdata = test_data, type = "response")
test_pred <- factor(ifelse(test_probs > 0.5, 1, 0), levels = c(0,1))
true_labels <- factor(test_data$high_vol, levels = c(0,1))

cm <- confusionMatrix(test_pred, true_labels, positive = "1")

# Print just the confusion matrix and accuracy
# Precision, Recall, F1, Sensitivity, Specificity
precision  <- cm$byClass["Precision"]
recall     <- cm$byClass["Recall"]
f1_score   <- cm$byClass["F1"]

print(cm$table)



library(pROC)
test_roc <- roc(
  response = true_labels,          # actual labels (0/1)
  predictor = test_probs,          # predicted probabilities
  plot = TRUE,                     # draw ROC curve
  print.auc = TRUE,                # print AUC on plot
  legacy.axes = TRUE               # optional: makes axes 1-specificity
)

# Extract numeric AUC
as.numeric(test_roc$auc)

library(forecast)
library(rugarch)
xreg_numeric <- data %>%
      select(cpi_diff, unemp_diff, fed_diff)

#Train / Test Split
set.seed(380)  # for reproducibility of any internal routines

n <- nrow(data)
split <- floor(0.7 * n)

train_vol <- data$spy_vol[1:split]
test_vol  <- data$spy_vol[(split + 1):n]
```

```r
train_x <- as.matrix(xreg_numeric[1:split, ])
test_x  <- as.matrix(xreg_numeric[(split + 1):n, ])

cat("TRAIN:", length(train_vol), "obs   TEST:", length(test_vol), "obs\n")

#ARIMAX model (volatility ~ own lags + macro differences)

spy_vol_ts_train <- ts(train_vol, frequency = 252)

arimax_model <- auto.arima(
      spy_vol_ts_train,
      xreg = train_x,
      seasonal = FALSE,
      stepwise = TRUE,
      approximation = FALSE
    )

cat("\n================= ARIMAX Model =================\n")
print(arimax_model)

# Forecast on test period
arimax_fc <- forecast(arimax_model, xreg = test_x, h = length(test_vol))$mean

arimax_rmse <- sqrt(mean((arimax_fc - test_vol)^2))
arimax_mae  <- mean(abs(arimax_fc - test_vol))

cat("\nARIMAX Model Performance\n")
cat("RMSE:", round(arimax_rmse, 6), "\n")
cat("MAE :", round(arimax_mae,  6), "\n")
cat("\n----------------- ARIMAX Coefficients -----------------\n")
arimax_coefs <- coef(arimax_model)
print(arimax_coefs)
cat("\n(Positive coefficients on surprises suggest higher volatility when surprises are positive;
negative coefficients suggest the opposite.)\n")

#Model Comparison + Coeffs
#GARCH(1,1) model on volatility (no xreg)
garch_spec <- ugarchspec(
  variance.model = list(
    model = "sGARCH",
    garchOrder = c(1, 1)
    ),
  mean.model = list(
    armaOrder = c(0, 0),
    include.mean = TRUE
    ),
  distribution.model = "norm"
  )

garch_fit <- ugarchfit(spec = garch_spec, data = train_vol)

cat("\n================= GARCH(1,1) Fit =================\n")
show(garch_fit)
# Forecast for test period
garch_fc <- ugarchforecast(
  garch_fit,
  n.ahead = length(test_vol)
)
```

```r
# conditional mean of volatility
garch_fc_values <- as.numeric(fitted(garch_fc))

garch_rmse <- sqrt(mean((garch_fc_values - test_vol)^2))
garch_mae  <- mean(abs(garch_fc_values - test_vol))

cat("\nGARCH Model Performance\n")
cat("RMSE:", round(garch_rmse, 6), "\n")
cat("MAE :", round(garch_mae,  6), "\n")

cat("\n----------------- GARCH(1,1) Coefficients -----------------\n")
garch_coefs <- coef(garch_fit)
print(garch_coefs)
cat("\n(Alpha + beta close to 1 indicates strong volatility clustering and persistence.)\n")
Model_Comparison <- data.frame(
  Model = c("ARIMAX", "GARCH(1,1)", "Linear Regression"),
  RMSE  = c(arimax_rmse, garch_rmse, rmse_val),
  MAE   = c(arimax_mae,  garch_mae, mae_val)
  )
cat("\n----------------- Model Comparison -----------------\n")
print(Model_Comparison)
cat("\n(Lower RMSE/MAE indicates better forecasting performance.)\n")
```